Robust Misspecified Models

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ROY Seminar, PSE & Sciences Po
Oct 13, 2025

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Much of literature focuses on a dogmatic agent who never changes her model

Questions

- Can a misspecified model persist in the long term if there are competing models?

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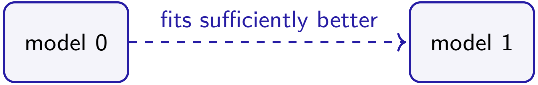
- Which misspecified models are more likely to persist than others?
- How do environmental factors contribute to persistence?

Model switching

I develop a new learning framework that allows for sticky model switching



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$$\boxed{\text{model } 0} \xrightarrow{\text{fits sufficiently better}} \boxed{\text{model } 1}$$

- Example: paradigm shifts (Kuhn, 1962)

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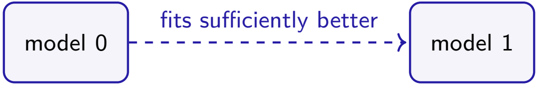

- Example: paradigm shifts (Kuhn, 1962)
- A middle ground between a dogmatic agent and an omniscient agent

Preview

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  Learning foundations for persistence of misspecified models
- **Main result**: general characterization of robust models with simple conditions

  Criteria to predict which underlying biases tend to persist in applications

Example: over vs underconfidence

built on Heidhues Köszegi Strack (2018)

Chooses how much effort to exert every day

$a_1, a_2, ... \in \{0, 1, 2\}$, cost of effort $= a_t(a_t + .5)$

Maximizes sales of paintings - cost of effort

sales $=$ $(a_t +$ talent$)$ $\times$ demand $+$ noise

where talent $= 1$, demand $= 2$

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- If Artist knows the true talent and demand, he will choose $a^* = 1$
- Artist has a non-degenerate prior about demand
- Artist has a potentially biased self-perception and assigns probability 1 to $\widehat{\text{talent}}$

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- Suppose Artist entertains a competing model with the correct talent
- Is this underconfidence going to persist? **No**

If Artist exerts an effort of 1:

$$\underbrace{(\hat{a} + \textcolor{green}{\mathsf{talent}}) \cdot \textcolor{blue}{\mathsf{demand}}}_{\text{predicted by competing model}} = (1+1) \cdot 2 = 4$$

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No effort choice is a self-confirming equilibrium $\Rightarrow$ model does not persist

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Result: Overconfidence persists if Artist's prior is relatively concentrated around $\widehat{\text{demand}}$

Lessons

- Seemingly similar biases may have different persistence properties

  Overconfidence $\Rightarrow$ **positive** belief reinforcement $\Rightarrow$ **high** asymptotic accuracy

  Underconfidence $\Rightarrow$ **negative** belief reinforcement $\Rightarrow$ **low** asymptotic accuracy

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- We need a full-fledged model-switching framework to characterize persistence

Framework

Primitives

In each period $t = 0, 1, 2, ...$, a myopic agent does the following:

- chooses an action a_t from a finite set $\mathcal{A}$ (e.g. effort)
- then observes an outcome y_t from set $\mathcal{Y}$ (e.g. sales)
- then obtains a flow payoff $u(a_t, y_t)$ (e.g. sales - cost of effort)

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True data-generating process (DGP) is fixed and unknown:

$$\text{outcome } y_t \text{ is drawn from probability density } q^*\left(\cdot|a_t\right)$$

Models

A model θ is a collection of predictions on how actions affect the outcome distribution

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where $\omega \in$ finite parameter space Ω^θ

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Examples. In Artist's model, parameter is market demand

In a linear supply model, parameters are slope and intercept

Models (cont.)

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Let θ^* denote the model that solely consists of the true DGP, and call it the true model

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- Let m_t denote the model choice in period t, where $m_0 = \theta$

Model θ : $y_t \sim q^{\theta}(\cdot|a_t, \omega)$, where $\omega \in \Omega^{\theta}$

True DGP: $y_t \sim q^*(\cdot|a_t)$

Model θ' : $y_t \sim q^{\theta'}(\cdot|a_t, \omega')$, where $\omega' \in \Omega^{\theta'}$

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$\boxed{\text{True DGP: } y_t \sim q^* (\cdot|a_t)}$

$\boxed{\text{Model } \theta : y_t \sim q^{\theta} (\cdot|a_t, \omega), \text{ where } \omega \in \Omega^{\theta}}$

switch between models θ and θ'

$\boxed{\text{Model } \theta' : y_t \sim q^{\theta'} (\cdot|a_t, \omega'), \text{ where } \omega' \in \Omega^{\theta'}}$

Operating within a model

Agent has full-support priors for each model:

$$\pi_0^{\theta} \in \Delta\Omega^{\theta}, \quad \pi_0^{\theta'} \in \Delta\Omega^{\theta'}$$

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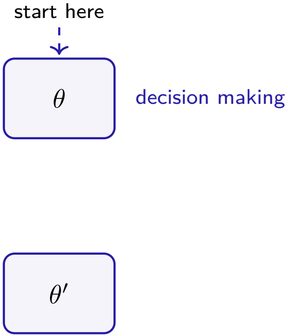

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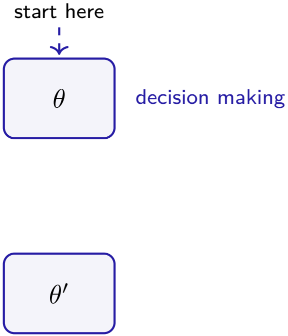

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- Agent calculates the Bayes factor,

$$\lambda_t = \ell_t(\theta')/\ell_t(\theta)$$

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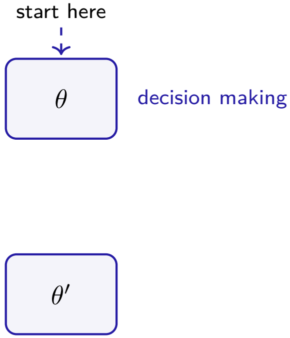

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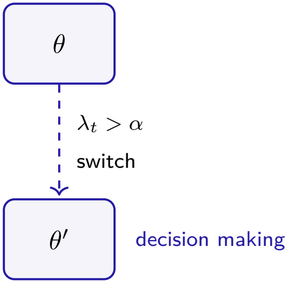

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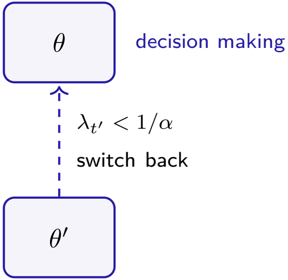

Bayes factor

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Bayes factor

data's likelihood given parameter

$$\lambda_t = \frac{l_t(\theta')}{l_t(\theta)} = \frac{\sum_{\Omega^{\theta'}} \pi_0^{\theta'}(\omega') \; \boxed{q^{\theta'}(y_0, \dots, y_{t-1} | a_0, \dots, a_{t-1}, \omega')}}{\sum_{\Omega^{\theta}} \pi_0^{\theta}(\omega) \; \boxed{q^{\theta}(y_0, \dots, y_{t-1} | a_0, \dots, a_{t-1}, \omega)}}$$

Bayes factor

prior-weighted sum data's likelihood given parameter

$$\lambda_t = \frac{l_t(\theta')}{l_t(\theta)} = \frac{\sum_{\Omega^{\theta'}} \boxed{\pi_0^{\theta'}(\omega')} \; \boxed{q^{\theta'}(y_0, \ldots, y_{t-1} | a_0, \ldots, a_{t-1}, \omega')}}{\sum_{\Omega^{\theta}} \boxed{\pi_0^{\theta}(\omega)} \; \boxed{q^{\theta}(y_0, \ldots, y_{t-1} | a_0, \ldots, a_{t-1}, \omega)}}$$

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- Common criterion in statistics

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- Common criterion in statistics
- Conceptual coherency with Bayes rule

Bayes factor in recursive form

$$\lambda_t = \underbrace{\boxed{\lambda_{t-1}}}_{\text{past evidence}} \times \overbrace{\frac{\sum_{\Omega^{\theta'}} \boxed{\pi_{t-1}^{\theta'}(\omega')} \boxed{q^{\theta'}(y_{t-1}|a_{t-1},\omega')}}{\sum_{\Omega^{\theta}} \boxed{\pi_{t-1}^{\theta}(\omega)} \boxed{q^{\theta}(y_{t-1}|a_{t-1},\omega)}}}^{\text{most recent evidence}}$$

- The Bayes factor λ_t measures *cumulative* balance of evidence supporting θ' relative to θ

Persistence

Three possible scenarios:

- m_t converges to θ
- m_t oscillates forever
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Observation: Persistence is prior-sensitive and relative to the competing model

Robustness

Definition. Fixing prior π_0^θ, model θ is:

globally robust at π_0^θ if θ persists against **every** $\theta' \in \Theta$ **at every** $\pi_0^{\theta'}$.

locally robust at π_0^θ if there exists $\epsilon > 0$ s.t.

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- Measure distance between θ and θ' by Hausdorff distance between the sets of DGPs
- Measure distance between π_0^θ and $\pi_0^{\theta'}$ by Prokhorov metric over distributions of DGPs

Preview of results

Theorem 1: which models are locally/globally robust for at least one full-support prior?

Theorem 2: at which priors are these models locally/globally robust?

First step

Consider a correctly specified competing model θ'

Q: Which models can persist against θ'?

First step

Consider a correctly specified competing model θ'

Q: Which models can persist against θ'?

A: Such a model has perfect asymptotic accuracy

It must induce the agent to play a self-confirming equilibrium in the limit

Equilibrium concept

Stability condition

Equilibrium concept

(Fudenberg & Levine, 1993)

Definition: Strategy σ is a self-confirming equilibrium (SCE) under model θ if

1. strategy σ is myopically optimal against some belief $\pi^\theta \in \Delta\Omega^\theta$
2. belief π^θ is consistent at strategy σ, i.e.

$$q^\theta(\cdot|a,\omega) \equiv q^*(\cdot|a), \forall a \in \mathrm{supp}(\sigma), \forall \omega \in \mathrm{supp}(\pi^\theta)$$

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a θ-modeler eventually only plays actions in $\mathrm{supp}(\sigma)$ with positive prob

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Lemma. A SCE is p-absorbing if it is quasi-strict

First step

Lemma: If θ persists against a correctly specified model, then θ admits a p-absorbing SCE.

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What's the implication?

- necessary condition for global robustness
- not directly related to local robustness

Which models can be locally/globally robust?

Theorem 1

The following are equivalent:

(i) model θ is locally robust for at least one full-support prior

(ii) model θ is globally robust for at least one full-support prior

(iii) there exists a p-absorbing SCE under model θ

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Proof idea

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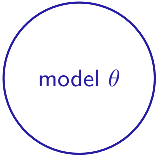

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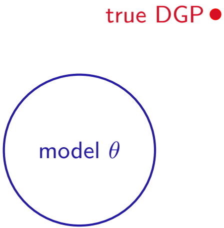

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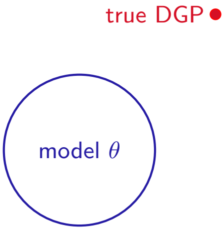

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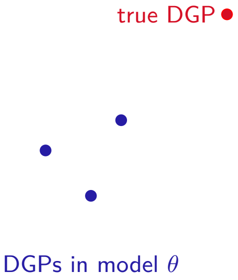

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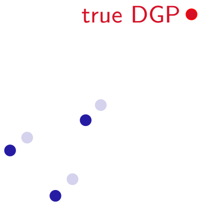

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DGPs in model θ'

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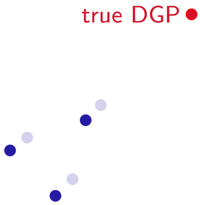

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persists against $\theta' \Rightarrow$ induces a p-absorbing SCE

Proof idea (cont'd)

Step 2: p-absorbing SCE implies global robustness at some prior

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p-absorbing $\Rightarrow$ reachable
self-confirming $\Rightarrow$ perfect accuracy

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Key step: $\exists$ prior sufficiently concentrated around SCE, s.t. prob. of being trapped < 1

Summary

	global	local
asymptotic accuracy	perfect	perfect
prior tightness	?	?

Summary

From now on, focus on models with no traps

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Insight: limiting step size does not expand the set of robust models

rather, it allows for more diffuse priors

Under which priors are models locally/globally robust?

Let C^θ be the set of parameters within θ associated with any p-absorbing SCE

- e.g. For overconfidence, $C^\theta = \{\widehat{\text{demand}}\}$; for underconfidence, $C^\theta = \emptyset$

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Suppose model θ has no traps,

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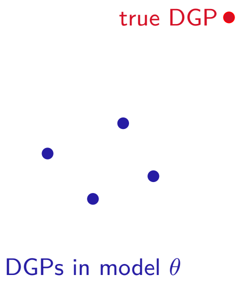
Want to show: global robustness $\Rightarrow \pi_0^\theta\left(C^\theta\right) \times \alpha \geq 1$



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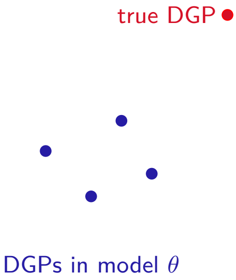

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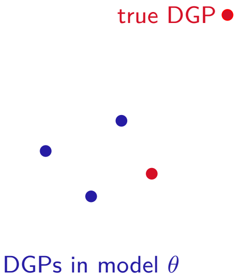

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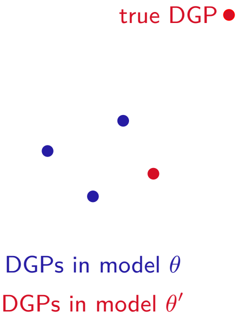

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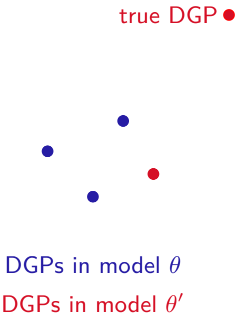

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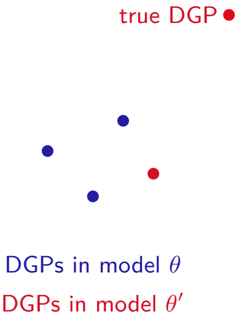

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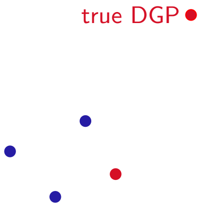

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Insight. "Simple + accurate", or "extreme + misleading"?

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- **Application.** A simplistic binary political view can outperform a more flexible correct view

Implication 2: lower switching stickiness

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  - Harder for misspecified models to persist... Good
  - But also harder for correctly specified models to persist... Not good

Extensions

Non-myopic agent

If non-myopic within each models, main theorems go through

Multiple competing models

An interesting overfitting issue arises if $\alpha <$ number of competing models

Alternative switching rules

Characterization depends on how data likelihoods are aggregated

Conclusion

Literature

Learning with a single misspecified model

- Specific biases: Nyarko (1991), Eyster & Rabin (2010), Rabin & Voyanos (2010), Epstein, Noor & Sandroni (2010), Ortoleva & Snowberg (2015), Bohren (2016), Spiegler (2016, 2019, 2020), Gagnon-Bartsch (2017), Fudenberg, Romanyuk & Strack (2017), Heidhues, Köszegi & Strack (2018), Mailath & Samuelson (2020), Ba & Gindin (2022), He (2022)
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Comparison of models

- Statistical fit approach: Cho & Kasa (2015, 2017), Gognon-Bartsch, Rabin & Schwartzstein (2023), Schwartzstein & Sunderam (2021), Olea, Ortoleva, Prat & Pai (2022)
- Objective payoff approach: Sandroni (2000), Blume & Easley (2006), Frick, Iijima & Ishii (2021), He & Libgober (2022), Fudenberg & Lanzani (2022)
- Subjective expected utility approach: Levy, Razin & Young (2022), Eliaz & Spiegler (2020)

What I do

Framework: misspecified learning with sticky model switching

What I do

Framework: misspecified learning with sticky model switching

Results: general characterization of robust models with simple conditions

What I do

Framework: misspecified learning with sticky model switching

Results: general characterization of robust models with simple conditions

	global	local
asymptotic accuracy	perfect	perfect
prior tightness	yes	no

Next steps

Many ways to utilize the characterization of robust models

- provide foundation for other forms of model misspecification
- evaluate particular biases across different decision environments
- shed light on the direction of new empirical research and policy making

Many new interesting questions within this model-switching framework ▶

Appendix